

Homework 4 - Solution

Student Number:

Name:

Problem 1. (30pt) Intel SGX threat models

Intel SGX assumes the operating system is not trusted. If the untrusted OS manages the page tables, what kinds of attacks can the OS do?

The operating system can map the physical page of an enclave to an unauthorized application.

It can also change mapping of the virtual page of an enclave to another physical page that is not supposed to be mapped.

Permissions to memory pages can be manipulated.

Problem 2. (30pt) Intel SGX and ARM TrustZone

Compare security features provided by Intel SGX and ARM TrustZone. Identify and explain two major differences.

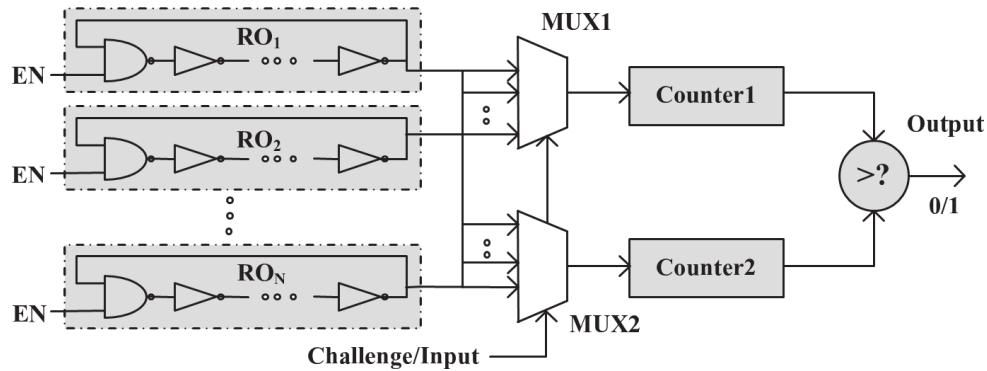
Intel SGX isolates processes while ARM TrustZone separates execution environments.

A single OS is employed for Intel SGX whereas ARM TrustZone employs a rich OS for Normal World and a separate trusted OS for Secure World.

The integrity of the trusted application is verified at the launch time in Intel SGX, but it is done at the booting time in ARM TrustZone.

Problem 3. (40pt) RO PUF

For the Ring Oscillator (RO) Physical Unclonable Function (PUF), explain the followings.



(a) RO PUF generates random outputs by comparing the two counters. Explain why the two counters can become different.

The output of an RO is an oscillating signal. A counter counts the number of edges of the connected RO output. The oscillating frequency depends on the delay of inverters in the RO. Due to the variance in delay, the oscillating frequency varies, resulting in different values of counters.

(b) There could be unstable ROs in the PUF. Explain how to improve stability of the PUF.

We can exclude unstable ROs to be selected by the challenge.

(c) When and how aging can affect the PUF output?

Aging slows down the frequency of RO outputs. If the frequency of two ROs drop at a different rate, and the frequency eventually crosses over, the output is flipped.